

7 — Random Sampling and Sampling Distribution

PROGRAM:

```
import numpy as np

import matplotlib.pyplot as plt

data = np.arange(1, 101)

sample = np.random.choice(data, size=10, replace=False)

print("Random Sample:")

print(sample)

means = []

for i in range(100):

    sample = np.random.choice(data, size=10, replace=False)

    means.append(np.mean(sample))

print("\nFirst 10 Sample Means:")

print(means[:10])

plt.hist(means, bins=10)

plt.xlabel("Sample Mean")

plt.ylabel("Frequency")

plt.title("Sampling Distribution of Sample Mean")

plt.show()
```

OUTPUT:

Random Sample:

[38 87 78 40 83 33 55 30 82 73]

First 10 Sample Means:

[np.float64(50.9), np.float64(52.3), np.float64(57.4), np.float64(52.7), np.float64(56.8),
np.float64(48.8), np.float64(56.1), np.float64(51.0), np.float64(45.0), np.float64(52.8)]

